Failure Analysis Report

FAIL-0518 — Assertion Failure

1. Failure Summary

Failure ID	FAIL-0518
Failure Type	Assertion Failure
Severity	Trivial
Component	Authentication
Application	APP-001 — SmartHome
Build Version	5.5.3
Timestamp	2024-05-27 08:37:26
Environment	ENV-006 — Cloud Test Env
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-0567
Title	Validate that data synchronization returns the correct result when two concurren...
Execution ID	EXEC-02636
Test Run	RUN-005 — Regression Cycle 5 — Build 5.1.6
Requirement	REQ-0264 — Reliability — Data Synchronization

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete health metric upload within the defined SLA.
Observed Behaviour	The system exhibited assertion failure when the device has been offline for more than 24 hours.

4. Error Details

Error Message:

```
AssertionError: expected 'connected' but received '404'.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.Diagnostics.execute(line 158)
```

Log Reference	LOG-94414

5. Executed Steps at Time of Failure

1	Navigate to Settings > Connectivity and tap 'Verify Identity'.	No crash or ANR is observed.

SYNTHETIC DATA — No real Samsung confidential information

2	Check server logs for event code 'EVT-5467'.	The fallback UI is displayed with a human-readable error message.
3	Assert that database record for entity 'ENT-978' has status='SYNC...	The fallback UI is displayed with a human-readable error message.
4	Launch the SmartHome application on the test device.	The performance counter logs a value below the SLA threshold.

6. Root Cause Analysis

Missing null-check on the device registry response before dereferencing the session token.

7. Linked Defect

Defect ID	BUG-0445
Title	Assertion Failure in Authentication: AssertionError: expected 'connected' but re...
Severity	Trivial
Priority	Low
Status	ASSIGNED
Root Cause	Missing null-check on the device registry response before dereferencing the sess...
Resolution	—
Assigned To	ENG-091

8. Historical Similar Failures

HIST-0105	FAIL-0106	Assertion Failure	4.3.3	2023-04-07	7	Resolved
HIST-0112	FAIL-0113	Assertion Failure	3.3.5	2023-12-13	14	Wont Fix
HIST-0301	FAIL-0302	Assertion Failure	3.5.2	2024-01-27	14	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information